

50 Logical Reasoning Questions in C Programming

Section 1: Output Prediction

Q1. `int a = 5;`
`printf("%d", a++ + ++a);`

Q2. `int a = 10;`
`if (a = 5)`
 `printf("True");`
`else`
 `printf("False");`

Q3. `int x = 10;`
`printf("%d", x == 10 ? 100 : 200);`

Q4. `int a = 5, b = 5;`
`if (a++ == ++b)`
 `printf("Equal");`
`else`
 `printf("Not Equal");`

Q5. `int x = 3;`
`x = x << 1;`
`printf("%d", x);`

Q6. `int a = 10;`
`int b = a++ + ++a;`
`printf("%d", b);`

Q7. `int arr[] = {1, 2, 3, 4};`
`printf("%d", *(arr + 2));`

Q8. `int x = 0;`
`while (x < 3)`
 `printf("%d", x++);`

Q9. `int x = 2;`
`printf("%d", x++ * x++);`

Q10. `int a = 5;`
`int b = 10;`
`printf("%d", a > b ? a : b);`

Q11. `char str[] = "Hello";`
`printf("%c", *str);`

Q12. `int a = 1;`
`int b = 1;`
`if (a-- && b++)`

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```
        printf("Yes");  
else  
    printf("No");
```

Q13. `int a = 5;`
`printf("%d", sizeof(a++));`

Q14. `int x = 1;`
`do {`
 `printf("%d", x);`
`} while (x-- > 1);`

Q15. `int a = 5;`
`int *p = &a;`
`*p = *p + 1;`
`printf("%d", a);`

Section 2: Code Logic Understanding

Q1. `int x = 0;`
`if (x = 10)`
 `printf("Yes");`

Q2. `const int x = 10;`
`x = 20;`

Q3. `int *ptr;`
`*ptr = 5;`

Q4. Can a function return an array in C?

Q5. `int a = 5;`
`{`
 `int a = 10;`
`}`

Q6. What happens if you access an array out of bounds?

Q7. `int i;`
`for (i = 0; i < 10; i++);`
 `printf("%d", i);`

Q8. `int a = 5;`
`printf("%d", ++a++);`

Q9. `int a = 1, b = 2;`
`int c = a+++b;`

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Q10. Which statement is true about ``static`` keyword?

```
Q11. int i = 0;
while (i++ < 5)
    continue;
printf("%d", i);
```

Q12. What is ``NULL``?

Q13. Which one is correct to declare a function returning pointer to int?

Q14. When does a segmentation fault occur in C?

Q15. Can a ``void`` function return a value?

Section 3: Pointers and Memory

```
Q1. int a = 5;
int *p = &a;
printf("%d", *p);
```

```
Q2. int *ptr = NULL;
printf("%d", *ptr);
```

Q3. What is the difference between ``malloc()`` and ``calloc()``?

Q4. What does ``free()`` do?

```
Q5. int *p;
*p = 10;
```

Q6. Can a pointer point to a function?

Q7. What is a dangling pointer?

Q8. How do you allocate a 2D array dynamically?

Q9. What happens if you ``free()`` memory twice?

```
Q10. int x = 10;
int *p = &x;
int **q = &p;
printf("%d", **q);
```

Section 4: Conceptual & Logical

Q1. Why is ``main()`` special in C?

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Q2. What is the output type of `sizeof()`?

Q3. `printf("%d", sizeof('A'));`

Q4. What are header files used for?

Q5. Difference between `==` and `=` in C?

Q6. What happens when `break` is used in a loop?

Q7. What is a function pointer?

Q8. Difference between `int func()` and `int func(void)`?

Q9. What does `volatile` mean?

Q10. `int a = 1;`
`int b = 0;`
`printf("%d", a && b || !a);`